

ICD-9-CM codes with no match in the target system

Descriptive analysis, 27 August 2026

There are 9 ICD-9-CM codes with no ICD-10-CA match and 52 with no ICDA-8 match. The reference crosswalk lists them with no partner, so the right answer for them is no mapping at all.

The algorithm uses two numbers to decide. How often the two codes appear together in patient records, and the cosine similarity between the two code labels. This describes both of those for each of these codes on its own, rather than as an average over the crosswalk. The similarity scores are ClinicalBERT.

ICD-9-CM to ICD-10-CA, the 9 codes

Co-occurrence

All 9 codes appear in the co-occurrence data. Partners is the number of different ICD-10-CA codes each one shares a record with, and the columns after it describe the counts across those partners. The source data does not report a count below 7, which is why every code has a minimum of 7.

Code	Label	Partners	Min	25th	Median	75th	Max	Most frequent partner
338	pain not elsewhere classified	827	7	18	49	181.5	13,431	E11, Type 2 diabetes mellitus
515	postinflammatory pulmonary fibrosis	306	7	13	24	53.8	5,194	J84, Other interstitial pulmonary diseases
239	neoplasms of unspecified nature	585	7	14	34	91	4,862	I10, Essential (primary) hypertension
327	organic sleep disorders	349	7	12	23	49	3,108	G47, Other sleep disorders
249	secondary diabetes mellitus	161	7	12	18	38	1,783	E11, Type 2 diabetes mellitus
209	neuroendocrine tumors	124	7	11.8	17	31.8	506	C78, Secondary malignant neoplasm of respiratory and digestive organs
175	malignant neoplasm of male breast	24	7	8	10.5	22.5	131	C79, Secondary malignant neoplasm of other and unspecified sites
339	other headache syndromes	58	7	9.2	15	17.8	54	E11, Type 2 diabetes mellitus
445	atheroembolism	2	7	9.2	11.5	13.8	16	I74, Arterial embolism and thrombosis

These counts are not low. Code 338 shares a record with 827 different ICD-10-CA codes, and its highest count is 13,431. Codes 515, 239 and 327 all reach the thousands. For comparison, across the 345 codes that do have a match the middle value of the highest count is 1,288.5, so this group sits slightly above that rather than below it.

The two exceptions are 445 and 339 at the bottom of the table. Those are the only two that look the way a code with no valid target was expected to look.

What the counts seem to reflect is how often a code is recorded, not whether it has a valid target. Pain, sleep disorders and unspecified neoplasms are recorded constantly, so they turn up next to whatever else

the patient has.

Cosine similarity

Each code is scored against all 2038 ICD-10-CA codes, so each row below describes 2038 numbers.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
175	malignant neoplasm of male breast	0.825	0.641	0.796	0.826	0.849	0.955	C30, Malignant neoplasm of nasal cavity and middle ear
249	secondary diabetes mellitus	0.817	0.663	0.774	0.817	0.859	0.951	O24, Diabetes mellitus in pregnancy
239	neoplasms of unspecified nature	0.831	0.662	0.804	0.833	0.859	0.942	D15, Benign neoplasm of other and unspecified intrathoracic organs
515	postinflammatory pulmonary fibrosis	0.852	0.694	0.823	0.858	0.886	0.932	N72, Inflammatory disease of cervix uteri
209	neuroendocrine tumors	0.836	0.693	0.808	0.842	0.867	0.915	A74, Other diseases caused by chlamydiae
327	organic sleep disorders	0.770	0.541	0.725	0.773	0.823	0.915	P78, Other perinatal digestive system disorders
339	other headache syndromes	0.816	0.648	0.786	0.820	0.851	0.915	A69, Other spirochaetal infections
338	pain not elsewhere classified	0.805	0.685	0.785	0.807	0.827	0.893	R52, Pain, not elsewhere classified
445	atheroembolism	0.711	0.552	0.686	0.713	0.736	0.853	M66, Spontaneous rupture of synovium and tendon

These scores are not low either. The highest score for these 9 runs from 0.853 to 0.955. Across the 345 codes that do have a match it runs from 0.861 to 0.981, so the two groups sit on top of each other.

The closest ICD-10-CA code is clinically unrelated in 8 of the 9 rows. Code 175, malignant neoplasm of male breast, comes out closest to C30, malignant neoplasm of nasal cavity and middle ear, at 0.955. Code 515, postinflammatory pulmonary fibrosis, comes out closest to N72, inflammatory disease of cervix uteri. The scores are as high as the ones the algorithm gets right, and the answers are not close.

The one exception is code 338, whose closest target is R52, and both are labelled pain not elsewhere classified. The reference crosswalk still records no match for it, so that one may be worth checking rather than assuming the algorithm is wrong.

ICD-9-CM to ICDA-8, the 52 codes

Co-occurrence

There is nothing to tabulate here. None of the 52 codes appear in the co-occurrence data at all. A code is only listed if it shares a record with at least one ICDA-8 code, so for these 52 there is no data rather than a low count.

This is why the algorithm returns no match for them, which is the answer we want. It gets there by requiring co-occurrence data before it will accept a pair. The same gap covers 32 of the 302 codes that do have a valid ICDA-8 match, so it stays silent on those as well.

Cosine similarity

Each code is scored against all 858 ICDA-8 codes.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
164	malignant neoplasm of thymus heart and mediastinum	0.813	0.584	0.776	0.810	0.846	0.985	142, malignant neoplasm of salivary gland
175	malignant neoplasm of male breast	0.816	0.578	0.782	0.816	0.848	0.984	174, malignant neoplasm of breast
237	neoplasm of uncertain behavior of endocrine glands and nervous system	0.821	0.594	0.785	0.822	0.855	0.984	238, neoplasm of unspecified nature of eye brain and other parts of nervous system
179	malignant neoplasm of uterus part unspecified	0.812	0.633	0.780	0.810	0.842	0.980	149, malignant neoplasm of pharynx unspecified
238	neoplasm of uncertain behavior of other and unspecified sites and tissues	0.814	0.626	0.777	0.811	0.845	0.977	237, neoplasm of unspecified nature of other genitourinary organs
585	chronic kidney disease	0.848	0.680	0.813	0.847	0.886	0.973	582, chronic nephritis
165	malignant neoplasm of other and illdefined sites within the respiratory system and intrathoracic organs	0.796	0.587	0.755	0.795	0.832	0.971	195, malignant neoplasm of illdefined sites
615	inflammatory diseases of uterus except cervix	0.855	0.631	0.828	0.863	0.890	0.971	621, other diseases of cervix
334	spinocerebellar disease	0.864	0.704	0.833	0.867	0.898	0.968	348, motor neurone disease
721	spondylosis and allied disorders	0.863	0.675	0.830	0.868	0.902	0.968	721, osteitis deformans
234	carcinoma in situ of other and unspecified sites	0.828	0.632	0.794	0.828	0.857	0.966	195, malignant neoplasm of illdefined sites
335	anterior horn cell disease	0.851	0.674	0.816	0.856	0.889	0.965	365, choroiditis
249	secondary diabetes mellitus	0.821	0.638	0.780	0.827	0.864	0.964	250, diabetes mellitus
712	crystal arthropathies	0.832	0.673	0.790	0.833	0.878	0.964	523, periodontal diseases
230	carcinoma in situ of digestive organs	0.839	0.616	0.808	0.840	0.873	0.963	189, malignant neoplasm of other and unspecified urinary organs
233	carcinoma in situ of breast and genitourinary system	0.820	0.583	0.785	0.822	0.859	0.962	184, malignant neoplasm of other and unspecified female genital organs
516	other alveolar and parietoalveolar pneumonopathy	0.868	0.686	0.849	0.873	0.896	0.961	517, other chronic interstitial pneumonia
417	other diseases of pulmonary circulation	0.826	0.697	0.796	0.825	0.858	0.960	447, other diseases of arteries and arterioles
617	endometriosis	0.864	0.698	0.831	0.865	0.901	0.958	582, chronic nephritis
576	other disorders of biliary tract	0.854	0.680	0.823	0.858	0.888	0.956	508, other diseases of upper respiratory tract
176	kaposi sarcoma	0.855	0.668	0.826	0.862	0.888	0.955	218, uterine fibroma

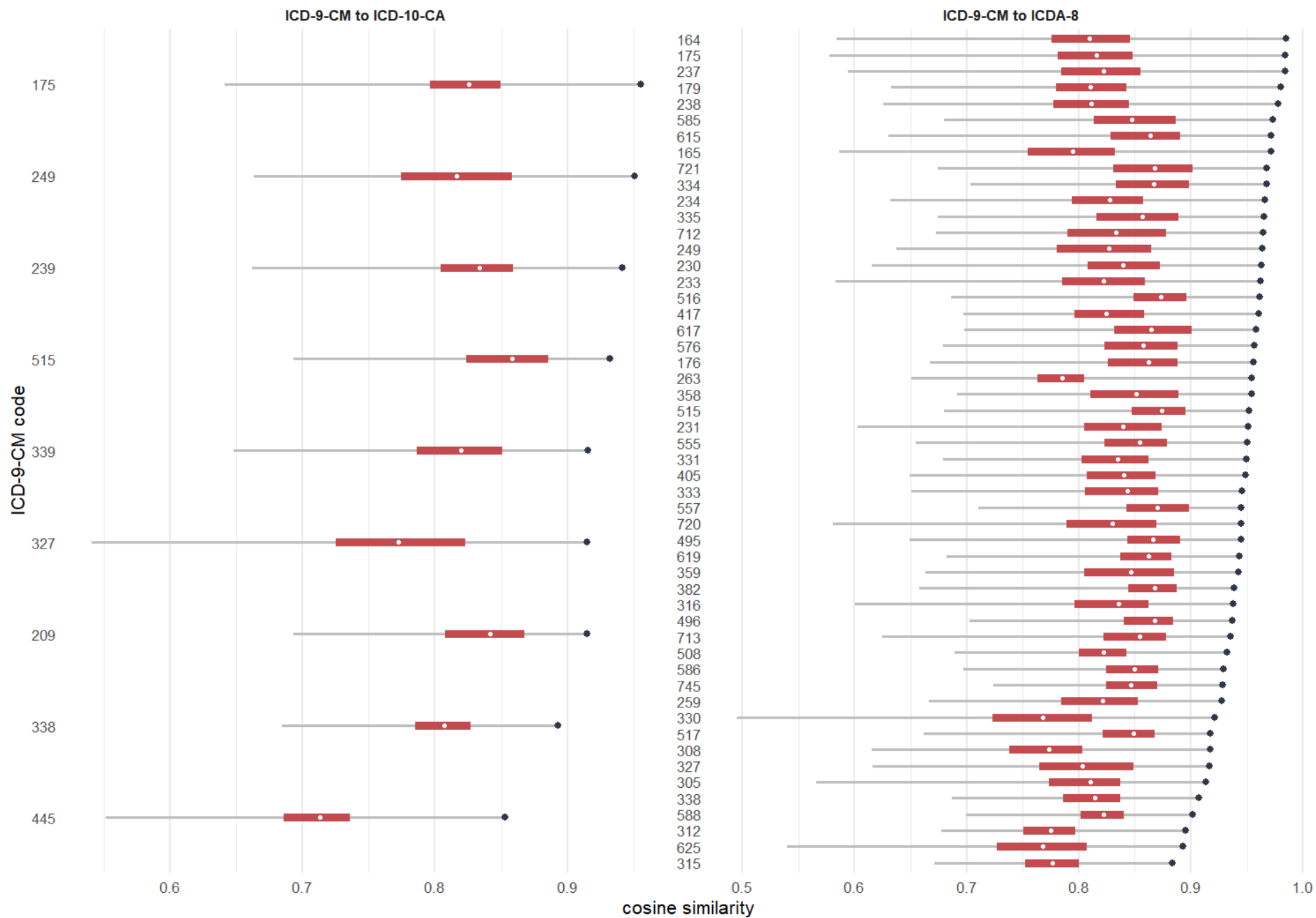
Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
263	other and unspecified protein-calorie malnutrition	0.783	0.651	0.763	0.785	0.805	0.954	267, protein malnutrition
358	myoneural disorders	0.849	0.691	0.810	0.851	0.889	0.954	523, periodontal diseases
515	postinflammatory pulmonary fibrosis	0.868	0.680	0.847	0.874	0.895	0.952	354, polyneuritis and polyradiculitis
231	carcinoma in situ of respiratory system	0.836	0.603	0.805	0.839	0.874	0.951	212, benign neoplasm of respiratory system
555	regional enteritis	0.851	0.655	0.823	0.854	0.879	0.950	422, acute myocarditis
331	other cerebral degenerations	0.832	0.679	0.803	0.835	0.862	0.949	344, other cerebral paralysis
405	secondary hypertension	0.836	0.649	0.807	0.840	0.869	0.949	401, essential benign hypertension
333	other extrapyramidal disease and abnormal movement disorders	0.835	0.651	0.805	0.844	0.871	0.945	360, conjunctivitis and ophthalmia
557	vascular insufficiency of intestine	0.867	0.711	0.842	0.870	0.898	0.945	424, chronic disease of endocardium
720	ankylosing spondylitis and other inflammatory spondylopathies	0.823	0.581	0.789	0.830	0.869	0.945	321, phlebitis and thrombophlebitis of intracranial venous sinuses
495	extrinsic allergic alveolitis	0.863	0.649	0.844	0.866	0.891	0.944	092, early syphilis latent
619	fistula involving female genital tract	0.856	0.682	0.837	0.862	0.882	0.943	939, foreign body in genitourinary tract
359	muscular dystrophies and other myopathies	0.842	0.663	0.805	0.847	0.885	0.942	733, other diseases of muscle tendon and fascia
382	suppurative and unspecified otitis media	0.863	0.658	0.844	0.868	0.888	0.939	612, acute salpingitis and oophoritis
316	psychic factors associated with diseases classified elsewhere	0.824	0.601	0.796	0.836	0.862	0.938	305, physical disorders of presumably psychogenic origin
496	chronic airway obstruction not elsewhere classified	0.862	0.703	0.841	0.868	0.884	0.937	584, renal sclerosis unqualified
713	arthropathy associated with other disorders classified elsewhere	0.844	0.625	0.822	0.855	0.878	0.935	775, hemolytic disease of newborn without mention of kernicterus
508	respiratory conditions due to other and unspecified external agents	0.821	0.689	0.800	0.822	0.842	0.932	515, pneumoconiosis due to silica and silicates
586	renal failure unspecified	0.847	0.697	0.824	0.850	0.871	0.929	583, nephritis unqualified
259	other endocrine disorders	0.818	0.667	0.784	0.821	0.853	0.928	117, other systemic mycosis
745	bulbus cordis anomalies and anomalies of cardiac septal closure	0.845	0.724	0.825	0.846	0.870	0.928	605, redundant prepuce and phimosis
330	cerebral degenerations usually manifest in childhood	0.761	0.495	0.723	0.768	0.812	0.921	095, other forms of late syphilis with symptoms
308	acute reaction to stress	0.769	0.616	0.738	0.773	0.804	0.917	307, transient situational disturbances
327	organic sleep disorders	0.804	0.617	0.765	0.803	0.849	0.917	717, other nonarticular rheumatism
517	lung involvement in conditions classified elsewhere	0.840	0.662	0.822	0.849	0.867	0.917	776, anoxic and hypoxic conditions not elsewhere classifiable
305	nondependent abuse of drugs	0.800	0.566	0.773	0.811	0.837	0.913	308, behavior disorders of childhood
338	pain not elsewhere classified	0.811	0.687	0.786	0.814	0.837	0.907	490, bronchitis unqualified
588	disorders resulting from impaired renal function	0.820	0.700	0.801	0.822	0.840	0.901	515, pneumoconiosis due to silica and silicates
312	disturbance of conduct not elsewhere classified	0.775	0.678	0.751	0.775	0.797	0.895	306, special symptoms not elsewhere classified
625	pain and other symptoms associated with female genital organs	0.762	0.540	0.727	0.768	0.807	0.893	781, other symptoms referable to nervous system and special senses
315	specific delays in development	0.777	0.671	0.752	0.776	0.800	0.883	066, late effects of viral encephalitis

The highest score for these 52 runs from 0.883 to 0.985. Across the 302 codes that do have a match it runs from 0.865 to 1.000. As with ICD-10-CA, the two groups overlap.

The same thing as a picture

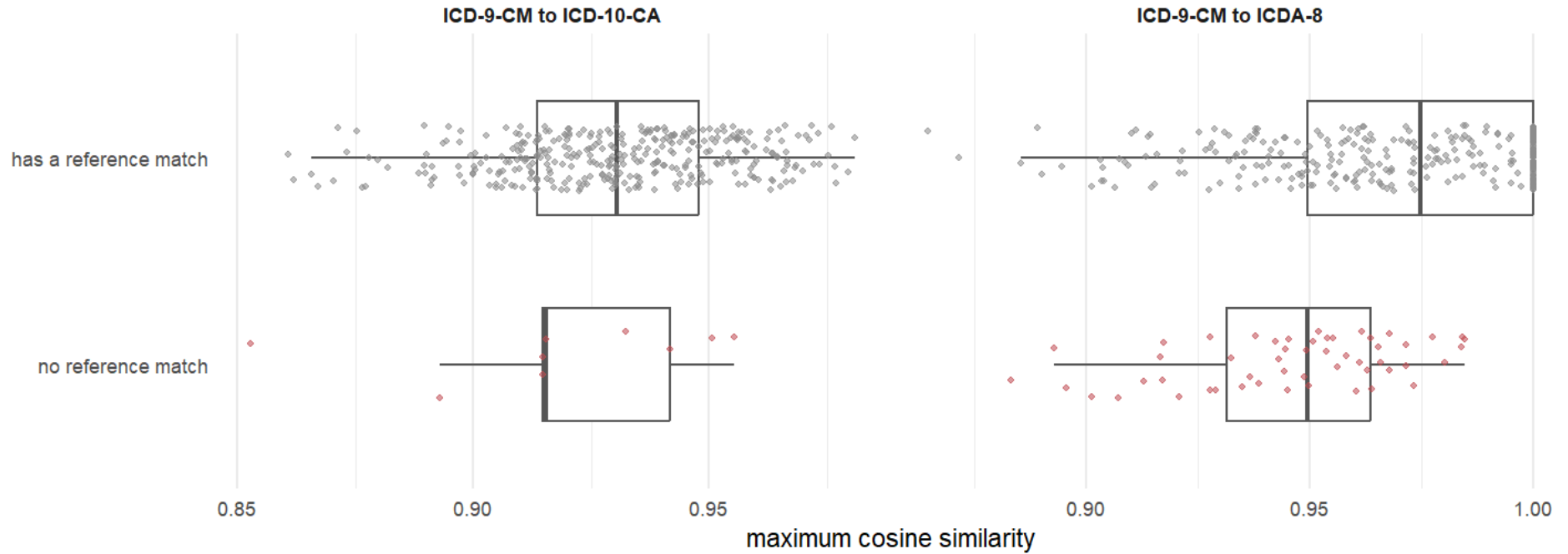
Each line is one code with no match. The grey line runs from its lowest to its highest similarity score against any target code, the red bar covers the middle half of its scores, the white dot is the median and the dark dot is the highest.

Cosine similarity of each unmatched ICD-9-CM code against every target code
 grey line min to max, red bar the quartiles, white dot the median, dark dot the maximum (ClinicalBERT)



The next one puts the codes with no match against the codes that do have one. Each point is a single ICD-9-CM code, placed at its highest similarity score.

Highest cosine similarity reached by each ICD-9-CM code one point per code, scored against every target code, ClinicalBERT



The scores for the codes with no match sit inside the range of the codes that do have one, on both crosswalks, so there is no cut-off on either signal that would separate them without also removing a large share of the mappings that are correct.